



TRUTH TABLE HELP



**You can resize this window by dragging its borders.
Please set the width so that the *'s both appear on the same line.**

The HELP facility is *not* intended to provide help with the substance of logic. A discussion of truth tables can be found in the text. Rules and definitions can be found in the References document (click the References button) in the module.

You can search this document to FIND a particular word or phrase by pressing **Ctrl+F**.

Quick Start

Use the Select button to select a problem from the program's database. (The program is also able to check problems created by a User. Click the User button.)

There are three kinds of problems in this module: "Truth Table Only" problems, which do not involve an argument and simply ask for a complete truth table for the given formula, problems that ask whether a given formula is a tautology, and problems that ask whether a given argument is tautologically valid.

Below the Problem line is the Question line (for example, "Is this argument tautologically valid?"), and below that there usually appear two tables side by side: the Setup table, which assigns "T"s and "F"s to atomic formulas, usually sentence letters, and the Main Table, which will assign "T"s and "F"s to the premises and conclusion of the argument. Some problems begin by asking you to fill out the Setup Table, in which the sentence letters are given their initial values. When you have done so, click "OK", and the Main Table will appear. Other problems provide a completed Setup Table.

Initially, all the cells in the Main Table are filled with "?". You must fill in a sufficient portion of the Main Table to determine the answer to the Question.

- For a Truth Table Only problem, you must fill in the entire column in the Main Table.
- If the Question asks whether the given formula is a tautology and you determine that the formula is *not* a tautology, you must indicate a row in which the formula is false by clicking in the white Check Box to the right of the row.
- If the Question asks whether the given argument is tautologically valid and you determine that the argument is *not* tautologically valid, you must indicate a row containing true premises and a false conclusion by clicking in the white Check Box to the right of the row.

To fill in a given cell in the Main Table, simply click on the cell. This will cause a Tree Window containing the parsed formula to open. The truth-values of the atomic formulas will already be filled in. Fill in the remaining truth-values by clicking on the truth-value button beneath the connective, and selecting "T" or "F". The References button at the

bottom of the main window brings up a document containing the rules for determining the truth-values of compounds from the truth-values of their parts.

Below the parsed tree is a Button-Row that mirrors the truth-value buttons in the tree. You can enter values using either the tree button or the button-row.

Tip: When you click on a truth-value button, hold the mouse button down, and drag the mouse pointer to the value you wish to select before releasing the mouse button.

Erroneous choices are called to the User's attention. When the assignment of truth-values is completed, click the OK button to insert the result into the Main Table.

For problems other than the Truth Table Only, when a sufficient portion of the Main Table has been filled in to determine the answer to the Question, answer it by clicking the button to the right of the Question. If the answer is "no", use a Check Box (to the right of the row) to indicate a row in which the premises are true and the conclusion is false. (If the answer is "yes", there should be no such row.)

Check your work by using the Check button or **Ctrl+k**.

Instructor Options

Some problems may require you to fill out the Setup Table yourself. You will be guided through this process. In a *correct* Setup Table, the right hand column alternates between "T" and "F", the next column between two "T"s and two "F"s, the next between four "T"s and four "F"s, and so on.

Some problems may require you to fill out a *complete* truth table rather than just enough to determine whether the argument is tautologically valid or not. If so, the problem will appear with a comment to this effect.

The text uses a traditional button-row style of display, and eventually you will need to be able to enter values using this style of display. In some problems, your instructor may have chosen to hide the parse tree, and to provide *only* the button-row for entry.

Some problems may be set to simulate exam mode. For those problems, most error messages will not appear unless you Check. (In an actual exam, Checking is disabled.)

Other Buttons

the User button allows you to enter a User problem by opening a edit window in which to create the problem. To create a problem that only asks that you fill out a complete truth table, check the "Truth Table Only" box. The usual facilities for entering symbolic expressions are enabled, including the Toolbox of logical symbols (invoked by a right-click). The standard Ctrl+Shift combinations for typing logical symbols are also available.

Ctrl+Shift+C	→	Conditional
Ctrl+Shift+B	↔	Biconditional

Ctrl+Shift+A	^	And
Ctrl+Shift+O	v	Or
Ctrl+Shift+N	~	Negation
Ctrl+Shift+T	∴	“Therefore”, used to separate premises from conclusion

Pressing **Enter** is equivalent to clicking the OK button.

Save your finished or unfinished work on a problem by clicking the Save button (or using **Ctrl+s**).

Do not fear the Delete button, it always first asks you to *confirm* that you wish to delete.

Submit your work over the Internet to the database of homework and exam submissions established for this course by clicking the Submit button. You will be asked to verify course information, and to enter your password, which you will create the first time you make a submission. Passwords are case sensitive. If you forget your password, you will not be able to submit your work. You may select more than one exercise to submit. The program will verify that the submitted work has been received. If a submission is unsuccessful, try it again. If consistently unsuccessful, report the error message to your instructor, and also send a Feedback message.

Print your work by clicking the Print button (or using **Ctrl+p**). The printout will contain your user information (name, Student ID#, etc.) as a header. (Note: At some institutions, you may need to first open a “printing account”.) The Print window allows you to select as many problems as you wish for printing. Only the problem plus “C” (for Correct) or “I” (for Incorrect or Incomplete) will be printed.

The Menu button brings the opening Menu module to the foreground, allowing you to select a different module or to open another instance of the present module. The Menu module also holds access to the course website, and the Quit button for closing the program. You will be asked if you wish to save your current work before quitting.

The Close button closes the present window. You will be asked if you wish to save your current work before the window is closed.

The Feedback button takes you to a web-based channel for communication with the Instructor, TAs, and authors of the program. It is especially useful for comments or questions about the program itself.